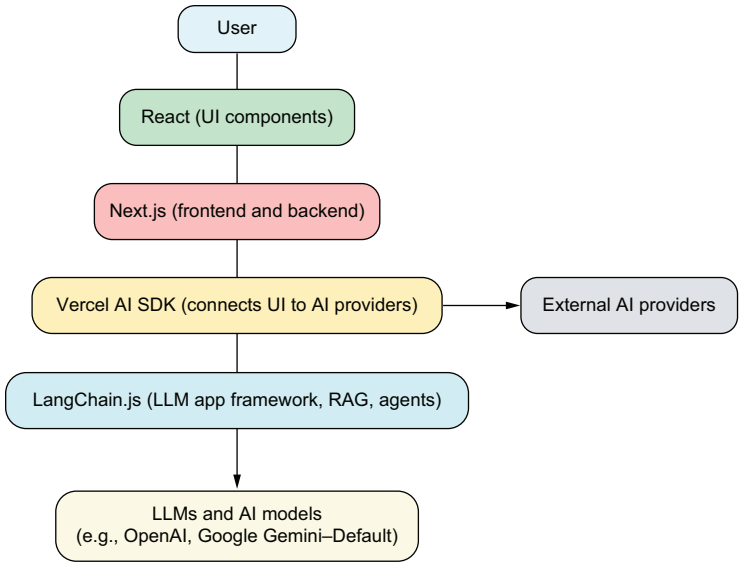


Figure 1.4 Using key technologies to create generative AI web applications

and data fetching, making development faster and more efficient. We will use Next.js to handle all the backend integrations.

The Vercel AI SDK is our cherry on top of the cake. It's an open source library specifically designed to integrate generative AI features into web applications, including specific hooks to handle state, components, and helpers that focus on core logic. We will use Vercel AI SDK tools to interact with external AI service providers and to incorporate AI functionalities seamlessly into our React components.

LangChain.js is a JavaScript framework for building applications with LLMs. It provides a suite of abstractions and building blocks that cover a wide range of application lifecycle scenarios from development to production. We will use LangChain.js to build conversational chatbots using RAG.

Other tools and frameworks for building AI web applications are available. Options include using an LLM API (e.g., the OpenAI.js client) directly or other small toolkits, such as polyfire-js and agentic tools, which offer similar capabilities. However, our choice of Vercel AI SDK has several advantages. First, it provides excellent integration with Next.js and React, allowing for seamless development of our chosen stack. The SDK offers good abstractions for handling multiple AI providers out of the box, saving developers from reinventing the wheel. Its convention-over-configuration approach simplifies setup and usage and offers a good state management solution (via UI and AI state abstractions). While developers could potentially write much of this functionality from scratch, doing so would be tedious.

The same argument applies to using LangChain.js, which offers additional advantages for building sophisticated AI applications. It seamlessly integrates with Next.js and